Lab 3 Instructions: ES 6 Classes and Git

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Objectives

The objectives of this lab are:

1. Practice using Git and GitHub.
2. Learn to write ES6 style classes and create and use objects from them.

Part 1: GitHub and Domino Classes

Set Up Your Git Repository

Create a GitHub Account

If you don't already have a GitHub account, create one now. As a student, you can get a free pro account plus other benefits by joining GitHub Education.

If you haven't already installed Git on your computer, download it and install it now.

Create a New GitHub Repository for Lab 3

1. Log In: Sign in to your GitHub account at github.com.
2. Start New Repository: Click the "+" icon in the top-right corner of any page and select "New repository".
3. Name the Project: Enter a short, memorable name for your repository in the Repository name field.
4. Set Visibility: Choose between Public (anyone can see it) or Private (only you and invited collaborators can see it).
5. Initialize (Optional): Check the boxes to add a README.md file for a project description or a .gitignore file to specify which files Git should ignore.
6. Finalize: Click the "Create repository" button.

   Create a Personal Access Token

   GitHub now requires users to use a *personal access token* to pull or push to their repository. This is a link to their instructions for setting that up. (BTW, if you create a "classic" token, the process is a little simpler.)

Clone the Repository

Clone the repository to your local computer:

1. Click on the green "Code" button.
2. Copy the HTTPS address for cloning your repository. (This address ends in `.git`)
3. On your computer, in a terminal app (or GitBash, PowerShell, etc.). At the folder where you want to clone the repository, type:
   `git clone addressofyourrepository`
   (Using the address you copied above.)

Refactor your Domino App

1. Add the latest version of your domino app to the local repository.
2. Modify your domino app to use classes.
   - See the Concentration game code for an example of this.
   - See the coding demo video (the link is on the LMS page) for examples and explanation.

Commit the Changes and Push them to GitHub

1. Stage Your Changes: Use `git add` to move your edited files into the Staging Area so Git knows which changes to include in your next snapshot.
   - Usage: `git add <file-name>` or `git add .` to stage all changes in the directory.
2. Commit the revised code: Use `git commit` to permanently record those staged changes to your Local History.

    - Usage: `git commit -m "Your descriptive message"`.
    - Note: Every commit requires a message explaining the update.
3. Push to the Remote: Use `git push` to upload your local commits to the Remote Server (GitHub).
    - Usage: `git push origin main`, or simply `git push` to push back to the default origin.
    - Impact: This moves your work from your local repository to the remote so others can see it.

After pushing, check your files on GitHub to make sure everything is there.
You can also check the status of your local repository with the command:
`git status`

Part 2: A Dice Game with Classes

Your instructor has provided you with a web app version of the dice game "Ship, Captain, Crew". The source code can be downloaded here. Use this code as an example to make a similar dice game. You should be able to re-use 90% of the code from the example. The dice game you make depends on the lab assignment version you are doing.

Note: You don't need to make your properties private or use getters and setters.

Version A: Going to Boston

You will make a simiplfied version of this game (Simplified becuase you will not keep track of rounds). Here are the rules you will implement in your game:

- Each player gets 3 rolls per turn.
- On each roll, you must keep (hold) the highest rolled die.
- After all three dice are kept (held), the sum of your dice is your score.
- Each player gets one turn.
- The player with the highest score wins!

Version B: Midnight

- Roll 6 dice. You must hold at least 1 die each roll.
- You must hold a 1 and a 4 to qualify.
- Your score is the sum of your remaining four dice.
- Each player gets one turn.
- Roll until all 6 dice are held. Highest score wins!

Version C: Threes Away

The goal is to get the LOWEST score possible.

3s count as zero (0). All other numbers count as face value.

You roll 5 dice and must hold at least one die per roll.

Once all 5 dice are held, your turn is over, and your score is the sum of the dice.

Each player gets one turn.

The player with the lowest score wins!

Submission

Beta Version and Code Review

- Share your part 2 beta version with a lab partner.
  - Your code should be at least 75% complete and working.
  - Share the code in your lab partner channel on Discord by sharing a link to your GitHub repository.
  - On Moodle, submit the link to your GitHub repository and report:
    - How complete was your web app, was it working?
    - Who did you share your beta code with?
    - What date and time did you share it?
- Do a code review for your lab partner.
  - Post the completed review form on Discord.
  - Submit your code review to Moodle.
- Revise your code and improve it based in part on feedback from your lab partner.

Production Version

- Upload your completed part 2 web app to citstudent.lanecc.edu. Test the app on the web server to verify that it still works there.
- Submit the following to Moodle:
  - Part 1 and 2
    - Put a link to your GitHub repositoryin the online text.

- Upload the code review from your lab partner with the "production" column filled out by you.
- Put the citstudent URL for your web app in the online text.

Note, some parts of these instructions were drafted using Gemini 3.

Written by Brian Bird, spring 2026.